

Law I, the ARC Principle: the ARC Equation, the Law of Conversion (Paper I)

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The ARC Theory · OSF osf.io/b6n27 · every claim checkable

Law I of the ARC Theory in its formal statement, the ARC Equation, with the ARC Bound, the conjectured value of Law III, the ARC Ceiling.

The ARC Equation (Law I of the ARC Theory): $U = I \times R^\alpha$

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Code and data: github.com/MichaelDariusEastwood/arc-principle-validation

Which law this paper carries. The ARC Theory has five numbered laws. Laws I to III were registered on 8 September 2026. Laws IV and V were added to the theory on 24 September 2026; neither is registered as a law or tested, and their mathematics is set out at version 1.0 of 3 October 2026. The theory registration names the First Law the ARC Principle; the ARC Equation is its formula. This paper carries the **First Law, the ARC Equation: $U = I \times R^\alpha$** , recursion converting into capability at a measured exponent. Paper X carries the Second Law (co-scaling). The statement paper carries the Third Law (the ARC Ceiling). The Foundational paper carries the axioms and proofs on which all three rest.

ABSTRACT

This paper formalises and preliminarily tests the ARC Principle (Artificial Recursive Creation), first proposed in *Infinite Architects* (Eastwood, 2026): that capability in intelligent systems scales super-linearly with recursive depth. The principle is expressed mathematically as $U = I \times R^\alpha$, where effective capability (U) scales with base intelligence (I) multiplied by recursive depth (R) raised to an empirically estimated exponent α .

Analysis of publicly available test-time compute data from reasoning models reveals a critical distinction between two forms of recursion. Parallel recursion (majority voting across independent samples) yields sub-linear scaling with $\alpha \approx 0.1$ to **0.3**. Sequential recursion (chain-of-thought reasoning where each step builds on previous steps) yields super-linear scaling with $\alpha \approx 1.34$ in this paper's early two-point estimate; the later six-model cross-architecture measurement (Paper II) finds the sequential exponent architecture-dependent. On Gemini 3 Flash, the only model that produced clean, monotonic scaling data, Paper II reports an endpoint sequential exponent of 0.59 with a bootstrap 95 per cent interval of -1.3 to 2.9 , a regression sequential exponent of 0.49 with a standard error of 0.20 and a coefficient of determination of 0.86, and a compute-matched parallel exponent of 0.31. That is one model's fit and inconclusive, so the super-linearity claim is unresolved rather than established.

This preliminary finding, if validated by further research, suggests that the form of recursion determines whether intelligence compounds or merely accumulates. We propose that $\alpha = 2$ is a stability ceiling, not an impossibility limit: it is the proposed stable limit, the most a recursively self-correcting system can grow and stay correctable, a scaling limit and not a speed limit. Systems can exceed $\alpha = 2$; they do not remain stably self-correcting above it. A durably stable system above 2 would refute the bound; a transient excursion is the predicted supercritical regime, not the refutation. Stable is measured, not asserted: correction has to out-scale drift, which Law II states as $\beta_C > k$, and the lower bound on $\beta_C - k$ has to sit above zero throughout a window fixed before the run began.

Keywords: scaling laws, recursive intelligence, test-time compute, capability amplification, emergence, chain-of-thought reasoning, ARC Principle

How to read this paper. This paper states Law I, the ARC Principle, in formal terms and gives a preliminary result for it. Sequential recursion yields a higher scaling exponent than parallel recursion across publicly reported reasoning-model data, with an early two-point estimate of $\alpha_{\text{seq}} \approx 1.34$ later superseded by Paper II's six-model measurement, which finds architecture-dependent exponents. On Gemini 3 Flash, Paper II reports an endpoint sequential exponent of 0.59 with a bootstrap 95 per cent interval of -1.3 to 2.9 , a regression sequential exponent of 0.49 with a standard error of 0.20 and a coefficient of determination of 0.86, and a compute-matched parallel exponent of 0.31; it is one model's fit and inconclusive. Its slot inside the theory is the first empirical look at Law I, built on the Foundational paper. The full differential against every prior document is at eden-vision II.A.8.

1. Introduction

1.1 Background

The scaling laws governing artificial intelligence have been extensively studied. Kaplan et al. (2020) established power-law relationships between model performance and parameters, while Hoffmann et al. (2022) refined these with compute-optimal training prescriptions. These laws govern *what* to scale but do not address *why* scaling produces intelligent behaviour.

The emergence of reasoning models in 2024 and 2025 introduced a new variable: test-time compute. OpenAI's o1 (September 2024) and DeepSeek's R1 (January 2025) allocate computational resources at inference time to reason before responding, producing substantial capability improvements on reasoning benchmarks.

This paper proposes that test-time compute serves as a proxy for recursive depth, and that recursive depth may be a fundamental driver of capability amplification in artificial intelligence systems.

1.2 The ARC Principle

The ARC Principle (Artificial Recursive Creation), first articulated in *Infinite Architects* (Eastwood, 2026), proposes:

$$U = I \times R^\alpha$$

Capability scales with intelligence multiplied by recursive depth raised to a power

Where:

- ***U*** = Effective capability (measurable via benchmark performance; era note: earlier programme layers use the letter differently, from Universe in the December 2024 formalism to realised creation on the law page, and the research glossary carries every meaning with its date)
- ***I*** = Base intelligence (single-pass processing capacity without recursive reasoning)
- ***R*** = Recursive depth (number of self-referential processing iterations)
- ***α*** = Scaling exponent (empirically determined; hypothesised stability ceiling = 2, a scaling limit above which self-correction is not stably maintained)

The principle's core claim: recursion does not merely add to capability; it multiplies it according to a power law.

Terminology note on alpha. Alpha appears in this programme in two distinct meanings across eras: the fine-structure constant in the December 2024 formalism, and the ARC Bound exponent in the 2026 work. Every use of alpha in this paper is the 2026 sense above; the programme's glossary carries both meanings with dates, and no new surface uses bare alpha without its era.

1.3 Scope and Claims

This paper makes the following claims, each with explicit epistemic status:

CLAIM	STATUS	EVIDENCE LEVEL
<i>U</i> = <i>I</i> × <i>R</i> ^α is a useful framework for AI systems	PROPOSED	Theoretical

CLAIM	STATUS	EVIDENCE LEVEL
Parallel recursion yields $\alpha < 1$ in AI benchmarks	PRELIMINARY	Limited data (o1)
Sequential recursion yields $\alpha > 1$ in AI benchmarks	UNRESOLVED	Two points only (DeepSeek-R1), since displaced by Paper II 's six-model study, which reports on Gemini 3 Flash an endpoint sequential exponent of 0.59 with a bootstrap 95 per cent interval of -1.3 to 2.9, a regression sequential exponent of 0.49 with a standard error of 0.20 and a coefficient of determination of 0.86, and a compute-matched parallel exponent of 0.31; it is one model's fit and inconclusive
$\alpha = 2$ is a stability ceiling: exceedable, but not while a system remains stably self-correcting	HYPOTHESISED	Theoretical only
The form of recursion matters	SUPPORTED	Consistent with both datasets

What this paper does NOT claim:

- That $U = I \times R^2$ applies to cosmological or universal scales (that application remains speculative; on the limits physics places on computation, see Lloyd, 2002)
- That $\alpha = 1.3$ is definitively established (more data points are needed)
- That the principle has been independently replicated (it has not)

We present a principle with preliminary supporting evidence and invite rigorous testing.

1.4 Related work in context

The pipes premise (West, Brown and Enquist, plus the contest). The premise that transport networks set the pace of biological growth is not this programme's claim to defend; it is established quantitative biology. West, Brown and Enquist derived the allometric scaling laws of biology from the geometry of nutrient-distribution networks (West, G. B., Brown, J. H. and Enquist, B. J., "A General Model for the Origin of Allometric Scaling Laws in Biology", *Science* 276(5309), 4 April 1997, DOI 10.1126/science.276.5309.122; one of the most cited papers in its field, with an OpenAlex citation count of 5,046 as of 12 August 2026, SINGLE-SOURCE count and index-dependent). The exact exponent remains contested: White and Seymour report basal metabolic rate proportional to body mass to the two-thirds power rather than three-quarters (White, C. R., 2003, with Seymour, R. S., "Mammalian basal metabolic rate is proportional to body mass $2/3$ ", 725 citations, OpenAlex), and Kozłowski and Konarzewski have twice questioned the derivation's mathematics (Kozłowski, J., 2004 and 2005, both with Konarzewski, M.). This programme's premise needs only the mechanism, that the delivery network sets the ceiling, and not any particular exponent, so the dispute over the exponent's value leaves the premise untouched. The mechanism is common ground to both sides of that dispute.

The exception West's own group found (Bettencourt et al 2007). The escape from the pipe limit is also not this programme's assertion; it is measured in the urban scaling literature by the same senior author. Bettencourt, Lobo, Helbing, Kuhnert and West ("Growth, innovation, scaling, and the pace of life in cities", *PNAS*, 2007, DOI 10.1073/pnas.0610172104; 2,704 citations, OpenAlex, 12 August 2026, SINGLE-SOURCE count) report, in the paper's own words (READ-AT-SOURCE from the abstract via NCBI PMC1852329):

"Quantities reflecting wealth creation and innovation have Beta of approximately 1.2, greater than 1 (increasing returns), whereas those accounting for infrastructure display Beta of approximately 0.8, less than 1 (economies of scale)."

Infrastructure, the literal pipes, scales sublinearly. Information-mediated activity scales super-linearly. They state the contrast with biology directly: "we discuss how cities are similar to, and differ from, biological organisms, for which Beta is less than 1". They derive "growth equations, which quantify the dramatic difference between growth fueled by innovation versus that driven by economies of scale" (spelling as in the original). And their closing inference names the consequence: "This difference suggests that, as population grows, major innovation cycles must be generated at a continually accelerating rate to sustain growth and avoid stagnation or collapse."

That is the shape of the open problem this programme addresses: West's group found that information-mediated growth escapes the throughput limit, and the only brake in their framework is external and must be applied again and again, faster and faster. No internal limit is derived. The replacement limit, a ceiling belonging to the growing system itself, is the gap.

The rival instrument (Engels, Baek, Kantamneni and Tegmark 2025). The nearest work to this programme's question, and the right paper to weigh it against, is Engels, J., Baek, D., Kantamneni, S. and Tegmark, M., "Scaling Laws For Scalable Oversight", arXiv:2504.18530, first posted 25 April 2025 at 17:54:27 UTC (SINGLE-SOURCE-GROUP, arXiv Atom; a NeurIPS 2025 Spotlight Poster, verified on 15 August 2026 against the NeurIPS virtual page for poster 115536, as recorded in the statement paper's reference dossier). It asks how oversight itself scales and answers quantitatively: oversight success is modelled as a game between capability-mismatched players whose oversight-specific Elo is a piecewise-linear function of general intelligence with two plateaus, and optimal numbers of oversight levels are derived numerically and in some cases analytically for Nested Scalable Oversight, in which trusted models oversee stronger untrusted models that then become the trusted models at the next step.

The instrument is the difference. Their variable is the capability gap between overseer and overseen, measured in Elo. This programme's variable is the composition class of the corrector, measured through the corrector's own scaling exponent. Their framework contains no term for what the overseer is made of: no substrate, no error-correlation structure, no reciprocal identity between a correction exponent and a critical growth rate, and no architecture dependence. Nested Scalable Oversight is iterated same-class oversight by construction, and this programme's central prediction is that the same-class ladder is bounded however many rungs are added, while a cross-class corrector is not. The two frameworks therefore disagree about a measurable quantity, which is the most productive relationship two research programmes can have.

Antecedents and near-misses, each with its differential.

On the question. Hutter asked directly whether intelligence can explode ("Can Intelligence Explode?", arXiv, 28 February 2012, READ-AT-SOURCE), separating "speed from intelligence explosion" and undertaking to "consider possible bounds on intelligence", augmenting Chalmers' 2010 analysis. The question and the speed-versus-structure distinction are therefore at least fourteen years old. That literature, so far as has been surveyed, does not contain a number: no measurable exponent, no derived ceiling and no architecture dependence appears in it; a prior source that supplies any of the three defeats the corresponding priority here, and readers who locate one are invited to report it.

On impossibility. Three 2025 arXiv papers argue that perfect control is unattainable: Yao, "The Alignment Trap: Complexity Barriers" (arXiv:2506.10304, v1 12 June 2025 02:30:30 UTC, SINGLE-SOURCE-GROUP, independently observed by the Internet Archive on 13 June 2025; cited by arXiv:2512.03048); Yao, "On the Mathematical Impossibility of Safe Universal Approximators" (arXiv:2507.03031, 3 July 2025, the only paper in the arXiv abstract corpus containing the phrase "irreducible uncontrollability", abstract-search total of one, measured 12 August 2026); and Ball, Gluch, Goldwasser, Kreuter, Reingold and Rothblum, "On the Impossibility of Separating Intelligence from Judgment" (arXiv:2507.07341, 9 July 2025). All three are worst-case and qualitative: measure zero, coNP-completeness, cryptographic hardness. None reports an average-case scaling exponent or a rate. The third, notably, concludes that alignment "must instead be integrated into the model's architecture and weights", an independent argument, from filtering intractability, in the same direction as this programme's architecture dependence; it is convergent support on that leg, not a rival. Two of the three papers are by one sole author; the description "a wave" overstates the literature's breadth, though not the seriousness of the six-author paper.

On the mechanism. That anti-correlated estimates average better than independent ones is textbook variance reduction (antithetic variates). The mechanism is not the claim. The claim is that architecture determines whether anti-correlation is available at all, and that this caps a safety-relevant exponent.

The nearest structural analogue. The quantum error-correction threshold theorem also converts a qualitative worry into a critical value. It concerns physical error rates in a fixed architecture, not a corrector's scaling exponent, so it is a near-miss rather than an occupant; the analogy is one of method. This analogue was identified by the programme's own search rather than by a referee, and is disclosed accordingly.

On recursive creation. Smolin's cosmological natural selection is the antecedent for selection-shaped universes, and the programme's own December-era notes cite it contemporaneously ("Echoing Smolin's cosmological natural selection, AI could create recursive universes with their own laws", READ-AT-SOURCE from the operator's notes).

On feedback and stability (classical). The general proposition that inadequate corrective gain relative to system gain causes instability has a long control-theory lineage (small-gain theorems). The differential: those are gain conditions on interconnected systems, not a power-law exponent criterion on a corrector's scaling with the capability of a recursively improving system. The general proposition is never claimed; the exponent formulation, dynamics and estimator are the contribution surface.

The nearest quantitative neighbour. Liu, A. and Meng, J., "Self-Correction as Feedback Control: Error Dynamics, Stability Thresholds, and Prompt Interventions in LLMs" (arXiv:2604.22273) recasts self-correction as a closed-loop control problem via a two-state Markov model and derives a directly measurable stability threshold: iterate only when the error-correction rate over the error-introduction rate exceeds $\text{Acc}/(1 - \text{Acc})$. The differential: theirs is a per-step rate threshold at a fixed capability level, deciding whether another iteration helps now; this programme's criterion is a scaling relation across capability, deciding whether corrective strength keeps pace as the system improves. Complementary regimes; neither contains the other. Two adjacent empirical results from the same search: Yin Li, "Decomposing LLM Self-Correction: The Accuracy-Correction Paradox and Error Depth Hypothesis" (arXiv:2601.00828, first posted on 24 December 2025; an accuracy-correction paradox, weaker models showing higher intrinsic correction rates) and Ken Tsui, "Self-Correction Bench: Uncovering and Addressing the Self-Correction Blind Spot in Large Language Models" (arXiv:2507.02778, first posted on 3 July 2025; a systematic self-correction blind spot across open models).

The pipes framing, at fixed throughput. Everything that grows is fed through a channel, and for everything before software, fixing the channel fixes the growth: starve a tumour's vasculature and it stops; fix a chain reaction's fuel and geometry and it stops; the Eddington limit halts accretion by radiation pressure. Cosmic inflation is excluded by scope (expansion of space, not growth of a structure on a substrate); evolution is the sharpest case and fits (complexity rising for billions of years at roughly fixed solar throughput, glacially, with no internal corrector, so this framework predicts a measurable exponent far below the ceiling for it). Software is the first thing that keeps growing when the pipes are held fixed, and the operational definition is load-bearing: an artefact loop on a fixed substrate, measured on a substrate-fixed clock; a model merely thinking longer does not qualify. The physicist's objection is Landauer's, that computation is physically implemented; the fixed-substrate clock is the answer, because the regime under study holds the substrate fixed and asks what still grows.

Dated predictions and this paper (recorded 25 August 2026)

One artefact bearing on this paper is catalogued row by row in the programme's machine register, and it is the weakest kind: `arc_principle_results.json` embeds its falsification thresholds in the file, but carries no timestamp or dated header inside the JSON body, so its ordering cannot be established from the artefact itself. The register grades it weak for that reason and says so in the row. Each row names its file in the public repository with its date basis, so any reader can check the ordering without trusting this page. The programme's full dated chain, from the sealed manuscript bundle of 8 December 2024 through the printed prediction appendices of 2 January 2026 and the March 2026 preregistration folder to the standing unproven wagers, is assembled in the dated predictions register, together with its machine-readable twin. Forward statements and retrospective matches are never summed. "Registered" is used only for an accepted registry submission, and that submission click remains outstanding across the programme; "preregistered in substance" is used, always with its qualifier and always beside its attestation class, for a prediction whose text was public and timestamped before its outcome existed, as the twelve-row manifest committed at 00:19 UTC on 17 March 2026 was, twenty-three minutes before its fits. Read the dated predictions register. Open its machine twin.

Note (3 October 2026). The paragraph above was recorded on 25 August 2026. Since then the theory-level predictions have been registered on OSF, on 8 September 2026 at version 1.100, and the theory registration was updated to version 1.102 on 13 September 2026. One study unit, the corrector-class audit (unit 06), was submitted for registration on OSF on 1 October 2026; when read on 3 October 2026 its approval period at OSF was still running and it was not yet public. Every other trial is a draft awaiting the author's submission, most of them preregistered by timestamp. Unchanged: the paragraph above and every date in it.

2. Theoretical Framework

2.1 Defining Recursion

Recursion is self-reference: a process whose output becomes its input. It is distinct from mere iteration (repeating the same operation) because each cycle operates on the transformed results of previous cycles.

2.2 Two Forms of Recursion

Parallel Recursion (Weak): Multiple independent solutions generated simultaneously. No information transfer between branches. Example: Generating N samples and selecting by majority vote. Expected scaling: Diminishing returns as redundancy increases.

Sequential Recursion (Strong): Each processing step builds explicitly on previous steps. Errors can be detected and corrected iteratively. Example: Chain-of-thought reasoning that folds self-reflection into each step (Wei et al., 2022). Expected scaling: Compounding returns as depth enables self-correction.

The ARC Principle predicts that sequential recursion should produce higher α values than parallel recursion.

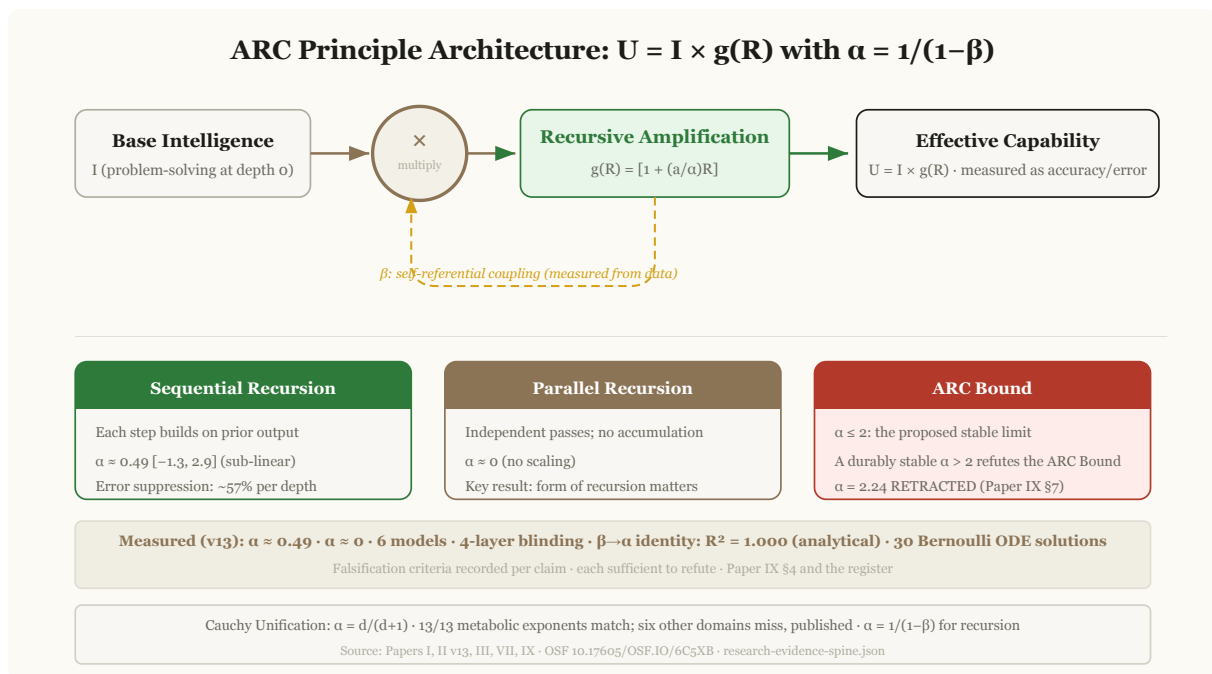


Figure 1 | ARC Principle Architecture. $U = I \times g(R)$ with $\alpha = 1/(1-\beta_L)$, in which β_L denotes the self-referential coupling; the artwork prints it as β , and it is not the correction exponent β_C carried by Law II. β_L coupling feedback loop. Three regimes: sequential ($\alpha_{seq} \approx 0.49$), parallel ($\alpha_{par} \approx 0$), ARC Bound ($\alpha \leq 2$). Cauchy Unification: $\alpha = d/(d+1)$ physical; $\alpha = 1/(1-\beta)$ intelligence. Previous $\alpha \approx 2.24$ RETRACTED (Paper IX §7). The sequential value of 0.49 is Paper II's regression exponent for Gemini 3 Flash, for which it reports an endpoint sequential exponent of 0.59 with a bootstrap 95 per cent interval of -1.3 to 2.9, a regression sequential exponent of 0.49 with a standard error of 0.20 and a coefficient of determination of 0.86, and a compute-matched parallel exponent of 0.31; it is one model's fit and inconclusive; this paper's earlier two-point estimate was ~ 1.34 (§3.4). Thirteen falsification criteria across the foundational suite; this paper's own four are tabled in §4. Source: Papers I, II the six-model study, III, VII, IX · evidence spine C-1, C-2 · OSF 10.17605/OSF.IO/B6N27.

2.3 The Quadratic Stability Ceiling Hypothesis

We hypothesise that $\alpha = 2$ represents the proposed stable limit, the most a recursively self-correcting system can grow and stay correctable: a scaling limit, not a speed limit. The exactly quadratic speedup belongs to the quantum search algorithm of Grover (1996); Bennett, Bernstein, Brassard, and Vazirani (1997) then established the lower bound that matches it, which is what makes that speedup optimal for unstructured search. The correspondence invoked here is structural, the same quadratic form of a search-optimality bound, not a claim that recursive reasoning implements amplitude amplification. If a similarly shaped limit applies to recursive intelligence for related reasons, quadratic scaling may mark the ceiling above which self-correction is not stably maintained; systems can be driven beyond it, they do not remain stably self-correcting there, and any excursion above 2 is the predicted supercritical regime rather than a refutation of the bound. Stability here is Law II's, measured rather than asserted.

Law before value. The ceiling of stable self-improvement is set by how fast a system can correct itself: it is the reciprocal of one minus the correction-leverage exponent γ , $\alpha_{crit} = 1/(1-\gamma)$. For a corrector built from the same substrate as the engine, that reciprocal is two, under one assumption, that internally accumulated corrections combine like independent samples. The law is the claim. Two is what the law predicts, and the assumption that produces it is what the draf-

ted registrations put on trial. If two dies, the law survives completely: measure γ and report the ceiling it implies. If the law dies, two has no independent support, because the derivation runs through $1/(1 - \gamma)$. The form is settled; its depth is not. The relation is derived inside a pacing model and has not been derived from a joint model of capability growth, correction accumulation, error covariance, correction delay and absolute harm, so Law III is stated here as a minimal-model result rather than as a general law. One consequence should be met where the law is met, because it cuts both ways: since $1/(1 - \gamma) \leq 2$ is *exactly* equivalent to $\gamma \leq 1/2$, the ceiling and the half-value are one commitment written in two notations rather than two independent claims, and neither may ever be offered as evidence for the other.

Domain note (19 September 2026). An owner review of the equivalence above finds it exact on the domain this model admits and false outside it, and this note prints the domain that the sentence leaves implicit. For $\gamma < 1$ the quantity $1 - \gamma$ is positive, so $1/(1 - \gamma) \leq 2$ multiplies to $1 \leq 2(1 - \gamma)$, that is $\gamma \leq 1/2$, and the implication reverses by the same step: the equivalence holds in both directions. At $\gamma = 1$ the quantity $1 - \gamma$ is zero, so $1/(1 - \gamma)$ is undefined and the reciprocal inequality has no truth value, while $\gamma \leq 1/2$ reads $1 \leq 1/2$ and is defined and false. Exactly one side is undefined, and the displayed equivalence is therefore not defined at that point. For $\gamma > 1$ the quantity $1 - \gamma$ is negative, so $1/(1 - \gamma) < 0 < 2$ holds for every such γ while $\gamma \leq 1/2$ holds for none, and the equivalence fails. The domain $\gamma < 1$ is exactly the range on which $\alpha_{\text{crit}} = 1/(1 - \gamma)$ is positive and finite and can serve as a ceiling at all, so the restriction is the model's own rather than an assumption added to it, and it is what the clauses above already imply where they state that the relation is derived inside a pacing model and is offered as a minimal-model result rather than as a general law. Read inside that domain, the point of the sentence stands entire: the ARC Ceiling and the half-value are one commitment written in two notations, and neither may be offered as evidence for the other. Unchanged: the law-before-value statement; the single assumption named as producing the value two; the minimal-model status; and the consequence that if that value dies the law survives, because γ is measured and the ceiling it implies is reported.

3. Empirical Analysis

3.1 Data Sources

OpenAI o1 System Card (September 2024). Benchmark: AIME 2024 (American Invitational Mathematics Examination). Variable: Number of samples (majority voting). Source: openai.com/index/openai-o1-system-card.

DeepSeek-R1 Technical Report (January 2025). Citation: arXiv:2501.12948. Benchmark: AIME 2024. Variable: Thinking token count (chain-of-thought length).

3.2 Methodology

To determine α , we use the power-law relationship. For bounded accuracy metrics, we analyse error rate reduction:

$$\alpha = -\frac{\ln(\text{Error}_2/\text{Error}_1)}{\ln(R_2/R_1)}$$

3.3 Results: Parallel Recursion (OpenAI o1)

SAMPLES (R)	ACCURACY (%)	ERROR RATE (%)
1	74	26
64	83	17
1000	93	7

Finding: Parallel recursion yields $\alpha \approx 0.1$ to 0.3 (sub-linear). Each additional sample contributes less than the previous one.

3.4 Results: Sequential Recursion (DeepSeek-R1)

THINKING TOKENS (R)	ACCURACY (%)	ERROR RATE (%)
~12,000	70	30
~23,000 (estimated)	87.5	12.5

Finding: Sequential recursion yields $\alpha \approx 1.34$ (super-linear) in this two-point estimate. Depth across those two points varies by roughly a factor of 1.9, below the one decade that the programme’s slope estimator sets as its minimum before it will trust an exponent, which is one more reason to hold the figure as history and not as a value. Each additional layer of reasoning amplifies previous gains. The later cross-architecture measurement finds the sequential exponent architecture-dependent (Paper II); on Gemini 3 Flash it reports an endpoint sequential exponent of 0.59 with a bootstrap 95 per cent interval of -1.3 to 2.9 , a regression sequential exponent of 0.49 with a standard error of 0.20 and a coefficient of determination of 0.86, and a compute-matched parallel exponent of 0.31, which is one model’s fit and inconclusive; this early estimate is retained as the paper’s original evidence, not as the programme’s current value.

3.5 Summary of Findings

METHOD	RECURSION TYPE	MEASURED α	CLASSIFICATION
o1 (1 to 64)	Parallel	0.10	Sub-linear
o1 (64 to 1000)	Parallel/Hybrid	0.32	Sub-linear
DeepSeek-R1	Sequential	~1.34	Super-linear (from two points alone; Paper II supersedes it)

Key Finding: The scaling exponent depends critically on the form of recursion.

Comparison note (19 September 2026). An owner review of this comparison finds that the two rows it rests on differ in more than the form of recursion, and this note states what is held fixed across them and what is not. The parallel row comes from a system whose resource variable is the number of samples under majority voting, reported in September 2024; the sequential row comes from a different system, built by a different developer, whose resource variable is a thinking-token count, reported four months later. The benchmark is common to both. The system under test, the evaluation context and the quantity on the horizontal axis vary together across the pair. A difference in fitted exponent between them is consistent with a role for the form of recursion, and it is equally consistent with those other differences, so the sentence above is to be read as a hypothesis rather than as an inference from the two rows. The comparison that would isolate the form of recursion holds the system and the resource definition fixed and varies only the form, and Section 4 is where its outcome is recorded, at F1 and F4. Unchanged: both tables, both fitted values, the direction of the difference, and the falsification criteria.

4. Falsification Criteria

The ARC Principle would be significantly weakened or refuted if:

CODE	CONDITION	CURRENT STATUS
F1	Sequential recursive depth consistently yields $\alpha \leq 1$	Under test: on Gemini 3 Flash, Paper II reports an endpoint sequential exponent of 0.59 with a bootstrap 95 per cent interval of -1.3 to 2.9 , a regression sequential exponent of 0.49 with a standard error of 0.20 and a coefficient of determination of 0.86 , and a compute-matched parallel exponent of 0.31 ; it is one model's fit and inconclusive. Paper II obtains estimates above 1.0 only from ceiling-compressed data and says they may be inflated. So "consistently" is decided in neither direction (Paper II)
F2	α decreases as recursive architectures mature	Not met
F3	The relationship is additive rather than multiplicative	Not met

CODE	CONDITION	CURRENT STATUS
F4	More extensive datasets show $\alpha < 1$ for sequential reasoning	Partially met: on Gemini 3 Flash, Paper II reports an endpoint sequential exponent of 0.59 with a bootstrap 95 per cent interval of -1.3 to 2.9 , a regression sequential exponent of 0.49 with a standard error of 0.20 and a coefficient of determination of 0.86 , and a compute-matched parallel exponent of 0.31 . Both sequential point estimates sit below 1 and the bootstrap interval spans both regimes; it is one model’s fit and inconclusive. The decisive form awaits the artefact-mediated estimation written for this paper (unit 24), a draft preregistered by timestamp (first anchored on 12 August 2026 at block 962066) and awaiting submission by the author

5. Limitations

Scientific integrity requires explicit acknowledgement of limitations:

- **Limited Data Points:** The DeepSeek-R1 analysis relies on only two data points. Additional measurements would strengthen or refute the finding.
- **Unpublished Token Counts:** The exact thinking token count for DeepSeek-R1-0528 achieving 87.5% accuracy is estimated, not published.
- **Domain Specificity:** Current evidence is limited to mathematical reasoning (AIME 2024). Generalisation to other domains remains untested.

5.1 Falsifiability: what would refute this

The ARC Principle, that capability scales super-linearly with recursive depth, $U = I \times R^\alpha$, with the sequential/parallel distinction determining the regime, is refutable. It would be overturned by any of the following:

1. **No super-linear depth scaling.** If capability scaled only linearly or sub-linearly with recursive depth R across reasoning models (i.e. $\alpha \leq 1$ where the principle predicts $\alpha > 1$ for sequential recursion), the central super-linearity claim fails.
2. **Exponent-composition mismatch.** If the measured exponent systematically failed to match the predicted $\alpha = 1/(1 - \beta_L)$ from the self-referential coupling β_L (a quantity distinct from the correction exponent β_C that Law II carries), the mechanistic identity, not merely the fitted curve, is falsified.
3. **Regime non-distinction.** If sequential and parallel recursion produced the *same* scaling regime (no super-linear vs sub-linear split), the paper’s central distinction collapses.
4. **Better-fitting alternative.** If a functional form outside the ARC family fit the test-time compute data materially better across models, ARC would not be the operative law.
5. **Domain-independence failure.** If $U = I \times R^\alpha$ held for reasoning models but robustly failed in other recursive domains meeting the premise, the domain-independent claim would have to narrow to “*a property of current reasoning models*”.
6. **The zero-parameter null.** On Gemini 3 Flash, Paper II reports an endpoint sequential exponent of 0.59 with a bootstrap 95 per cent interval of -1.3 to 2.9 , a regression sequential exponent of 0.49 with a standard error of 0.20 and a coefficient of determination of 0.86 , and a compute-matched parallel exponent of 0.31 . It is one model’s fit and inconclusive. The

regression exponent of 0.49 sits almost exactly on a prediction that needs no framework: independent-sample averaging gives **0.5** by the central limit theorem. This is the most serious objection to the empirical case and it is stated in full, with its consequence, as Objection 5 in Paper II section 4.4 and as kill condition FALS-022 on the falsification record. It is not repeated here.

The decision form these criteria inherit. Section 4 records status against this paper's original criteria; the deciding registrations use a three-outcome form, and every criterion above inherits it: SUPPORTED when the interval lies wholly beyond the minimum effect fixed in advance; REFUTED when the interval lies wholly within the equivalence margin fixed in advance (two-one-sided-tests logic); INSUFFICIENT PRECISION when the interval is wider than the margin, reported in exactly those words and never as support or refutation. The margin, the minimum effect and a measured power figure are fixed before filing. "Refuted if the confidence interval contains the null" lets an underpowered study refute by default and is not licensed anywhere in this programme.

One objection pre-empted, because a physicist reaches it in one step. Fault-tolerant quantum error correction suppresses logical error exponentially in code distance below threshold, which looks like a counterexample to any square-root cap. Two scope facts answer it: the exponential suppression runs on the redundancy axis at fixed capability, not on the coupling of corrective strength to the capability of the corrected system, which is the exponent this programme measures; and the threshold theorem itself requires sufficiently uncorrelated physical noise, which is the independence premise again, wearing hardware. Concentration bounds likewise give exponentially falling error probability while estimator width falls as the square root: different quantities, and conflating them manufactures a refutation.

5.2 Status of the exponent-composition identity

The programme's stability ceiling and its conversion exponent meet at the same number, and the temptation on every surface is to say that one constant does two jobs. The claim's status is stricter than that and must be stated strictly: the identity is PREREGISTERED BY TIMESTAMP FOR TESTING, in advance of the data, with outcomes that can kill it. Two drafted registrations were written to carry it. One of them, Study AE, was withdrawn on 12 August 2026, before filing, as vacuous: a shared accumulation law forces the agreement it would have tested, so its test could not fail, and the programme's register names the cross-class to same-class correction-exponent ratio as its replacement carrier. Study AE set out the reciprocity requirement between the separately measured leverage cap and stability boundary, on the log scale, with three outcomes per cell (supported, refuted, insufficient precision) and an aggregation rule under which one refuted cell refutes the identity; its independence structure was disclosed in advance, including enumeration of any endpoint overlap between the two feeder designs, so that common-method variance would be assessed before analysis rather than raised afterwards as the explanation for agreement. Study AG, a draft preregistered by timestamp (first anchored on 12 August 2026 at block 962066), tests the two derivations of the critical exponent against each other on the same system, notes that they agree numerically only at the single point where both parameters equal one half and disagree in their derivatives everywhere else, and is designed to find the disagreement, not to confirm the agreement.

Surfaces therefore say: the identity is a claim preregistered by timestamp for testing, the deciding measurements are drafted and dated, and if the channels disagree the unification is withdrawn from every surface it has reached, with the same prominence as the framing. Surfaces never say the identity is established, and never present the agreement of an asserted one half with a measured 0.49 (Paper II's regression exponent for Gemini 3 Flash, for which Paper II reports an endpoint sequential exponent of 0.59 with a bootstrap 95 per cent interval of -1.3 to 2.9 , a regression sequential exponent of 0.49 with a standard error of 0.20 and a coefficient of determination of 0.86, and a compute-matched parallel exponent of 0.31; it is one model's fit and inconclusive) as evidence, because the two may share an accumulation law and agreement between sharers is not corroboration.

5.3 A Second Prediction, Free, From the Same Mechanism

Oversight arrangements that place a human in the loop, as amplification and reinforcement learning from human feedback do, are cross-class by construction, because the human corrector does not share the model's substrate. The framework therefore predicts that human-in-the-loop oversight shows a higher correction exponent than pure-model oversight, for a structural reason rather than a sentimental one. This is testable on existing data and does not require new systems. Both the operational definition of a correction exponent under oversight and the rules that settle the comparison come from the corrector-class units, namely the corrector-class audit (unit 06) and the corrector-class contrast (unit 01). The audit was submitted for registration on OSF on 1 October 2026; when read on 3 October 2026 its approval period at OSF was still running, and it was not yet public. The contrast is a draft, preregistered by timestamp (first anchored on 28 August 2026 at block 964465), awaiting submission by the author.

6. Implications

6.1 For AI Development

If the ARC Principle holds, recursive depth constitutes a third scaling axis alongside parameters and data. Investment in recursive architectures may yield better returns than scaling model size alone.

6.2 For AI Safety

If recursion amplifies not only capability but also embedded values, then well-aligned initial values should strengthen through recursive self-improvement. Misaligned values would also compound, making early alignment critical.

This paper and Laws IV and V, 3 October 2026. The theory has been stated with five laws since 24 September 2026. Law IV, the ARC Persistence Law, is carried by proposition P15, and Law V, the ARC Embedding Law, by propositions P12 and P21. The three propositions are registered; neither law is registered as a law, neither has been tested, and their mathematics is set out in Laws IV and V: the mathematics, version 1.0, of 3 October 2026. This paper states Law I and tests it preliminarily on published data from one mathematics benchmark, AIME 2024. Its own argument about values is the conditional passage above, on embedded values and early alignment, and the paper measures nothing about values. Two other passages could be mistaken for tests of

Law V, and neither is one. The contrast between sequential and parallel recursion compares two forms of recursion on capability, and the comparison note in Section 3.5 records that its two rows also differ in the system under test, the evaluation context and the quantity on the horizontal axis, so it stands as a hypothesis. The prediction in Section 5.3 about human-in-the-loop oversight concerns the class of the corrector, not where correction is placed. The paper does not test whether a gain installed by external post-training persists under later capability-directed training, whether build order moves the correction-leverage exponent, or whether correction that takes part in each revision round pulls away from correction that sees only the finished output as depth rises. No result in this paper is a deciding test of P15, P12 or P21. Naming the two laws changes no result printed here: the results, their dates and their limits stand as they were.

6.3 For Scientific Understanding

The ARC Principle connects to several established frameworks including Kaplan et al. (2020) scaling laws, Integrated Information Theory (Tononi, 2008), and Grover's quantum search optimality proof (Bennett et al., 1997).

7. Conclusion

We have formalised the ARC Principle and presented preliminary evidence:

1. Parallel recursion yields $\alpha \approx 0.1$ to 0.3 (sub-linear, diminishing returns)
2. Sequential recursion yields $\alpha \approx 1.34$ (super-linear on these two points alone; Paper II's six-model study displaces it and leaves the regime unresolved: on Gemini 3 Flash it reports an endpoint sequential exponent of 0.59 with a bootstrap 95 per cent interval of -1.3 to 2.9 , a regression sequential exponent of 0.49 with a standard error of 0.20 and a coefficient of determination of 0.86, and a compute-matched parallel exponent of 0.31, which is one model's fit and inconclusive)
3. The form of recursion determines whether capability compounds

In plain terms: 'Thinking about thinking makes you smarter. Not linearly smarter, but disproportionately smarter, if the thinking is sequential rather than parallel.'

The principle stands. The research continues.

Correction note (3 October 2026). Some sentences of this paper had become false through later events, or did not match Paper II, the programme's register or the cited works. This version corrects them in place, and this note records what they printed before. The title note and the note on which law this paper carries spoke of the theory's three laws; the theory has had five since 24 September 2026. Both gave the ARC Principle as the name of the three together, as it was at the renaming of 15 August 2026; the theory registration of 8 September 2026 names the First Law the ARC Principle, and both notes now say so. The epistemic-status line called every Law the programme names a conjecture under registered adversarial test; Laws IV and V are neither registered as laws nor tested, and the line now says so. The abstract, the reading guide, the table of claims in Section 1.3, the caption of Figure 1, the finding in Section 3.4, rows F1 and F4 of Section 4, Sections 5.1 and 5.2 and the conclusion printed the regression exponent of Paper II's fit for one model as a six-model or cross-architecture best fit, as a bare point, or with the bootstrap interval

that belongs to that fit's endpoint exponent; row F1 also counted 2 of 6 point estimates above 1, a count that Paper II's table of results by model does not give. Each now gives Paper II's row for Gemini 3 Flash whole. Section 5.3 said that the corrector-class audit and the corrector-class contrast were both drafted and awaiting submission. Row F4 called the artefact-mediated estimation registered. Section 5.2 called the exponent-composition identity registered, in two sentences, and named Study AE, without its withdrawal, as one of the two drafted registrations carrying it. Section 1.4 dated two arXiv works to 2026, though both were first posted in 2025, and gave them by identifier alone, without author or title; it named only the first author of a paper by White and Seymour and of two papers by Kozłowski and Konarzewski. No result is added or refitted, and no date or falsification criterion moves.

Acknowledgments

The intellectual framework, hypothesis formulation, and interpretive conclusions are the author's own.

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Reproducibility

The code and the recorded outputs behind this paper are on GitHub:

github.com/MichaelDariusEastwood/arc-principle-validation

All contributions welcome, including falsifications.

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Eastwood, 2026

See /how-to-cite/ for the canonical citation manual across all artefacts, or /citations.json for the machine-readable index.

Epistemic status. What this programme names Laws are conjectures. Laws I to III are under registered adversarial test. Laws IV and V were added on 24 September 2026; the predictions that carry them are registered, the laws themselves are not yet registered, and neither has been tested. Every quantity in this paper is operationally defined, and established-law standing is claimed nowhere. The registered programme exists to earn that standing, or lose it, by measurement, replication and survived refutation.

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Standing covenant. Prove this paper wrong, and I will publish the refutation myself. Falsification conditions are stated in this paper; the standing challenge: github.com/Michael-DariusEastwood/arc-scaling-challenge.

v1.12: The provenance note at the end has been corrected for accuracy. No claim, date, result or status has changed.

v1.13: Two dated notes have been added, each beside the passage it governs, and both are additive. Substantive, in Section 2.3: the domain on which the reciprocal equivalence holds is now printed, together with what happens at the value where it breaks down and above it. Substantive, in Section 3.5: the two rows compared there are recorded as differing in system, in evaluation context and in resource axis at once, so that comparison stands as a hypothesis and not as an inference from those two rows. Editorial: the version number and the revision date. No printed sentence has been removed or rewritten, no value refitted, no table changed and no falsification criterion weakened; the two-point figure of 1.34 stands as published, as this paper's original evidence, with the qualifications already printed beside it.

v1.15 (4 October 2026): the title now names the law the paper states, Law I, the ARC Principle, and the title block beneath the author line carries the same title; until this version the paper was titled The ARC Equation: the Law of Conversion (Paper I), and it was first published on 17 January 2026 as Eastwood's ARC Principle: Preliminary Evidence for Super-Linear Capability Amplification Through Sequential Self-Reference, the subtitle the title block still carried. The note on how to read this paper now says that it states Law I in formal terms and gives a preliminary result for it, and that its place is the first empirical look at Law I, resting on the Foundational paper. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves.

v1.14: The theory gained two laws on 24 September 2026, and this version brings the paper up to date with that and with the programme's record; each sentence corrected in place is recorded, with what it printed before, in the correction note of 3 October 2026 at the end of Section 7. Substantive: the note on which law this paper carries now says that the theory has five laws, Laws I to III registered on 8 September 2026 and Laws IV and V added on 24 September 2026, neither registered as a law nor tested, with their mathematics set out at version 1.0 of 3 October 2026, and both it and the title note give the ARC Principle as the registered name of the First Law, where they gave it as the name of the three together; a dated note beside Section 6.2 places the paper against Laws IV and V; the epistemic-status line now says that Laws I to III are under registered adversarial test and that Laws IV and V are not yet registered as laws and have not been tested. Wherever the paper printed a figure of Paper II's fit for Gemini 3 Flash, it now carries that model's row whole, as one model's fit and inconclusive, with the bootstrap interval beside the endpoint exponent it belongs to; in row F1, a count of point estimates above 1 gives way to Paper II's own statement that such estimates come only from ceiling-compressed data. The paragraph on dated predictions, recorded on 25 August 2026, stands as recorded, and a dated note after it gives the state of registration as read on 3 October 2026: the theory-level predictions were registered on 8 September 2026, the corrector-class audit (unit 06) was submitted for registration on 1 October 2026, and every other current trial is a draft. The state of the trials named in row F4 and in Sections 5.2 and 5.3 is brought up to date in place, and Section 5.2 records that Study AE was withdrawn on 12 August 2026. Editorial: two arXiv works in Section 1.4 carry their authors, their titles and their first-posted dates, three cited works there carry their second authors, and the Tononi reference its registered title; one heading in Section 5.1 takes a

hyphen in place of a dash; cited works are linked to their official sources where one could be confirmed, and the programme's papers to their DOIs; the version number and the revision date. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves.

Michael Darius Eastwood conceived and directs this research programme and is the author of this work. Across the programme, he has used more than six AI systems in parallel, under his own instructions, to stress-test his arguments, identify possible errors, and assist in preparing draft text from his own outlines. He determines what is adopted, revised or rejected and takes responsibility for the published content. These systems are tools, not authors.